12 Probability: conditioning and weighted trees

Sheet 65 Tue 4 May 2027 *Probability model and equally likely outcomes*

1. Warm-up · Quick A fair coin is tossed three times. How many outcomes are there?
2. Core · Method Biased die: $P(6) = 0.25$ and the other five faces are equally likely. Calculate $P(1)$.
3. Core + · Critical thinking "It will rain or it will not rain tomorrow: so there is a one in two chance." Is this equally-likely model justified?
4. Review · Quick Which shapes make up the net of a square-based pyramid?
5. Thread · Translate Each side of a square of side x is increased by 3. Express the increase in area in terms of x.

Sheet 66 Mon 10 May 2027 *Law of large numbers (informal version): simulation*

1. Warm-up · Quick In 100 tosses, 47 are "heads". What is the relative frequency of "heads"?
2. Core · Interpret Why, in 10 tosses of a fair coin, can the frequency of "heads" stray from 0.5? What happens with 10,000 tosses?
3. Core + · True/false True or false? Justify. Tossing a fair coin 1,000 times gives exactly 500 "heads".
4. Review · Problem Price including tax $P(x) = 1.2\,x$ (20% VAT). Calculate $P(250)$ and the preimage of 480.
5. Thread · Method Solve in $\mathbb{R}$: $\dfrac{2x-1}{3} = \dfrac{x+4}{2}$.

Sheet 67 Tue 11 May 2027 *Python simulation of a repeated random experiment*

1. Warm-up · Quick $P(A) = 0.35$. Calculate $P(\overline{A})$.
2. Core · Reading code What does this program compute? Roughly what value will it display?

```
from random import randint
compte = 0
for i in range(1000):
    if randint(1, 6) == 6:
        compte = compte + 1
print(compte / 1000)
```

3. Core + · Method Fair 12-sided die with faces numbered 1 to 12. Probability of getting a prime number?
4. Review · Construction M lies on the perpendicular bisector of $[AB]$. What can be said about MA and MB? How is it constructed with compasses?
5. Thread · Recognise What is the shape of the cross-section of a cube by a plane parallel to one of its faces?

Sheet 68 Thu 13 May 2027 *Conditional probability $P_A(B)$*

1. Warm-up · Quick Among 25 canteen users, 15 are girls. A canteen user is chosen at random: probability that this is a girl?
2. Core · Method In a class of 30 pupils, 12 play football, of whom 5 also play handball. A pupil is chosen at random: calculate $P_F(H)$, where F: "plays football", H: "plays handball".
3. Core + · Method In an equally likely model, $\text{Card}(A) = 50$ and $\text{Card}(A \cap B) = 20$. Calculate $P_A(B)$.
4. Review · Quick Put in increasing order: $\dfrac{7}{4}$, $-\dfrac{3}{2}$, 0.5.
5. Thread · True/false True or false? Justify. A quadrilateral with two opposite sides parallel is a parallelogram.

Sheet 69 Tue 18 May 2027 *Conditional probability and two-way tables*

1. Warm-up · Quick In the following table, what is the total of the "sport" row?

	Girls	Boys	Total
Sport	18	22	…
No sport	12	8	20
Total	30	30	60

2. Core · Method A pupil is chosen at random from this table. Calculate $P(S)$, $P_F(S)$ and $P_S(F)$ (F: girl, S: does sport).
3. Core + · True/false True or false? Justify. In this table, $P_F(S) = P_S(F)$.
4. Review · Quick Calculate $5^0 + 2^{-1}$.
5. Thread · Method $ABCD$ is a parallelogram with $A(1,1)$, $B(5,2)$, $C(6,5)$. Find D.

Sheet 70 Thu 20 May 2027 *Weighted trees: construction*

1. Warm-up · Quick What is the sum of the probabilities of the branches leaving a single node?
2. Core · Method In a population, 30% have a disease M. A test is positive (T) for 90% of those with the disease and for 5% of those without it. Complete the branch probabilities of the tree.
3. Core + · Translate Translate into symbols: "among those with the disease, 90% test positive".
4. Review · Quick Calculate $\dfrac{2}{3} \div \dfrac{4}{9}$.
5. Thread · Problem Three lighthouses flash every 6, 8 and 9 seconds. They light up together at $t = 0$. When is the next simultaneous flash?

Sheet 71 Mon 24 May 2027 *Weighted trees: multiplying along a path*

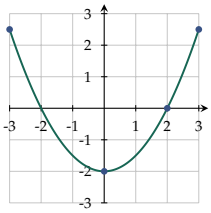
1. Warm-up · Quick Calculate 0.3×0.9.
2. Core · Method Using the tree from sheet 70, calculate $P(M \cap T)$ and $P(\overline{M} \cap T)$.
3. Core + · Method Deduce $P(T)$.
4. Review · Factorise Factorise $x^2 - 9$.
5. Thread · Quick The side $[BC]$ of length 13 cm is the hypotenuse of a right-angled triangle ABC. What is the radius of its circumscribed circle?

Sheet 72 Tue 25 May 2027 *From table to tree and symbolic notation*

1. Warm-up · Quick Express $\dfrac{30}{60}$ as a probability.
2. Core · Method From the table on sheet 69, build the tree (F or G, then S or $\overline{S}$): give $P(F)$, $P_F(S)$ and $P_G(S)$.
3. Core + · Translate Write symbolically and calculate "the probability of being a boy who does not do sport".
4. Review · Quick A car travels at constant speed: 150 km in 2 h. What distance in 5 h?
5. Thread · Statement → formula A taxi charges a pick-up fee of 2,000 Ar then 1,500 Ar per km. Express the price $p(x)$ for x km, then calculate $p(8)$.

Sheet 73 Thu 27 May 2027 *Reversing conditioning: $P_A(B)$ and $P_B(A)$*

1. Warm-up · Quick What is the difference between $P_M(T)$ and $P_T(M)$?
2. Core · Method Using the data from sheet 70 ($P(M \cap T) = 0.27$ and $P(T) = 0.305$), calculate $P_T(M)$.
3. Core + · Method Rare disease (1% of the population); the test is positive for 99% of those with the disease and 5% of those without. Calculate $P_T(M)$.
4. Review · Graph Curve of f:



   Read $f(2)$ and the preimages of 2.5.
5. Thread · Method Program: choose a number; multiply it by -2; add 10; divide by 2. Result for 7? Expression in x?

Sheet 74 Mon 31 May 2027 *False positives and false negatives*

1. Warm-up · Quick Define "false positive" and "false negative".
2. Core · Method 10,000 people, 2% with the disease; the test is positive for 90% of those with the disease and 10% of those without. How many false negatives, false positives and positives in total?
3. Core + · Interpret Using the previous data, calculate the proportion of people who truly have the disease among the positives and comment.
4. Review · True/false True or false? Justify. The area of a square is proportional to the length of its side.
5. Thread · Disprove Conjecture: "for every integer n, $n^2 + n + 41$ is prime". Test $n = 40$.